

Requirement Gathering and Analysis

This first stage of a software development life cycle focuses on the objective and requirements of the end user for whom the software has to be developed. This is a crucial phase in the life cycle since the software would be developed further basis on the requirements or the information gathered in this phase. This phase includes the analysis of functional and financial feasibility along with identifying the requirements from the stakeholders and defining the same in Software Requirement Specification (SRS). This SRS is then used in the later phases.

Design

Based on the requirements documented in the Software Requirement Specification (SRS) document, an architectural design is proposed for the software. This design is then documented in a design document. This phase includes identifying the hardware and software system based on the requirement and designing the system architecture. Creating UML (unified modelling language) diagrams such as use cases, class diagrams, sequence diagrams and activity diagrams are a part of this phase.

Develop

This is a phase where the software is actually developed and built and is usually the longest phase in the software development life cycle. In this phase, the actual code is written and is verified against the requirements by performing unit testing.

Testing

In this phase, the system is tested as a whole and the bugs/defects found in the system are reported, tracked and fixed. Different types of testing are performed until the system reaches the desired quality standards.

Deploy & Maintenance

Once the system is thoroughly tested in the testing phase, it is ready to go live. In this phase, the system is ready to be delivered, installed and put into use for the end user.

This phase also includes correction of the defects/errors that were not caught in the earlier phases along with the complete maintenance of the system which includes improving and ultimately enhancing the system.